Inverters with n-type MOSFET Load

* Inverter Circuit with a nmos active load instead of a load resistor

* The active load used in nmos inverter may be

  (i) Enhancement nmos Load

  (ii) Depletion nmos Load

Advantages of MOSFET as a Load:

ⓐ Silicon area → Less than transistor

ⓑ Overall performance is better compared to resistive load.

Enhancement Load NMOS Inverter

→ Load is made up of enhancement nmos.

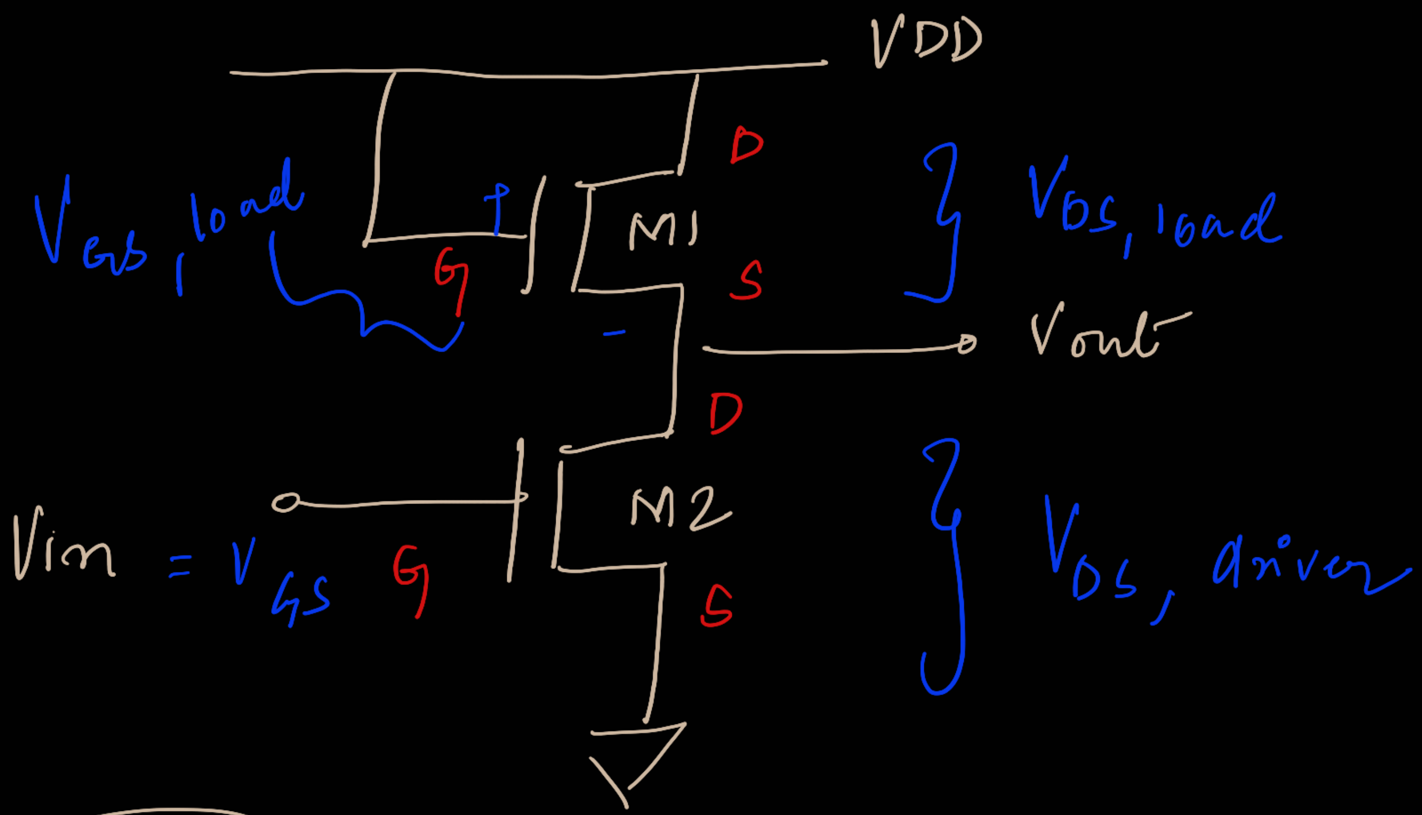
→ Two possible topologies of enhancement load nmos inverter

ⓐ Saturated enhancement type

ⓑ Linear enhancement type

Saturated Enhancement type

The nmos is in Saturation



M1 = Always in the ON Condition

Assume $VDD = 5V$, $V_T = 1V$

$$V_{GS} = V_G - V_S$$
$$= 5 - V_{out}$$

$$V_{DS} = V_D - V_S$$
$$= 5 - V_{out}$$

Load nmos is always in

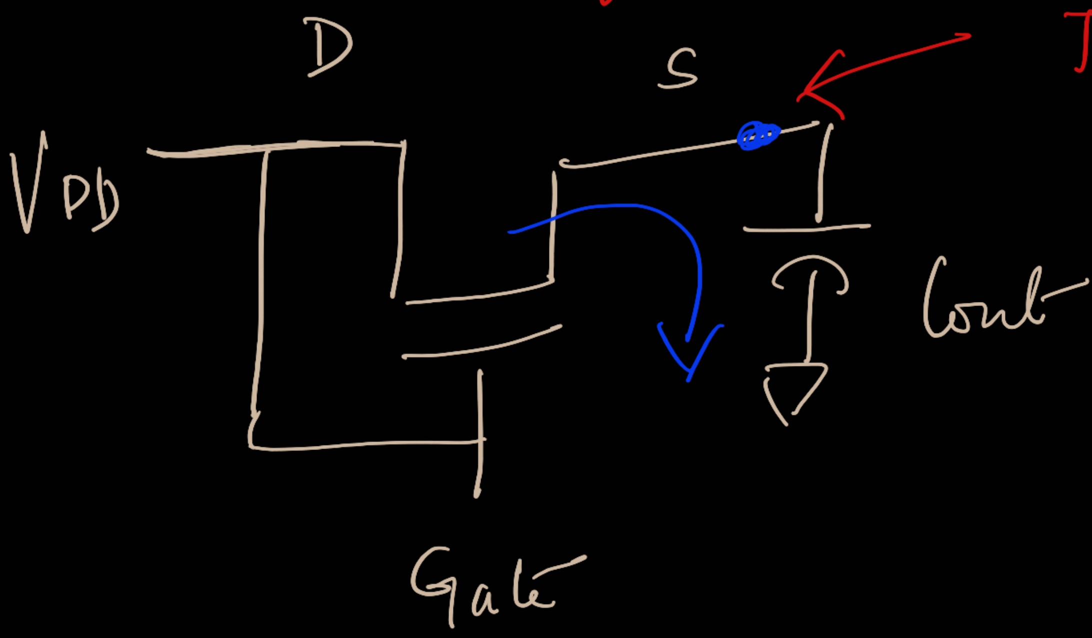
saturation

$$V_{DS} > V_{GS} - V_T$$

$$5 - V_{out} > \underbrace{5 - V_{out} - 1}_{4 - V_{out}}$$

Why V_{OH} level is limited to $V_{DD} - V_{T,load}$?

(Concept of Pass Transistor)



The source point starts to increase till the point

$V_S = V_{DD} - V_{T,load}$

$$V_{GS} = V_G - V_S$$

$$= V_{DD} - (V_{DD} - V_{T,load})$$

$$= V_{DD} - V_{DD} + V_{T,load}$$

$= V_{T,load}$

As a result, when source $> VDD - V_{T,load}$

$V_{GS,load} \le V_{T,load}$

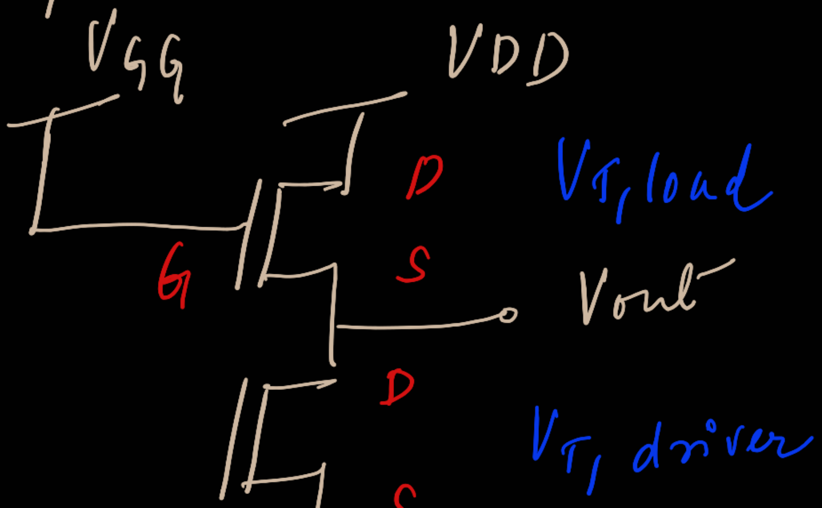
$\therefore$ The device will go to cut-off

DISADVANTAGE

The maximum $V_{OH} = VDD - V_{T,load}$

Linear Enhancement Load :

In this inverter, enhancement - nmos load is biased such that the load operate in linear region



$$\boxed{V_{GG} > V_{DD} + V_{T,load}}$$ ← Extra bias

① $V_{GS} > V_T$ ← nmos is always ON

② $\boxed{V_{DS} < V_{GS} - V_T}$

$V_{GS} = V_{GG} - V_S$

$V_{GG} = V_{DD} + V_{T,load}$

$= 5 + 1$

$V_{GG} = 6V$

$V_S = V_{DD} = 5$

$V_{GS} = 6 - 5$

$= 1$

$$\boxed{V_{GS} \geq V_{T,load}}$$

$V_D = 5\,V$ and $V_S = 5V$

$V_{DS} = 0V$

$V_{DS} < V_{GS} - V_{T,load}$

$V_{DS} = 0$

$V_{GS} - V_{T,load} = 0$

$V_{DS} \leq V_{GS} - V_{T,load}$

$$\boxed{\text{nmos} = \text{Linear}}$$

$$V_{GS} = V_{GG} - V_S$$
$$= V_{DD} + V_{T,load} - (V_{DD} - V_{T,load})$$
$$= \cancel{V_{DD}} + V_{T,load} - \cancel{V_{DD}} + V_{T,load}$$
$$= 2V_{T,load}$$

When $V_S = V_{DD}$

$$V_{GS} = V_{DD} + V_{T,load} - V_{DD}$$
$$= V_{T,load}$$ (The Device is conducting)

The device is capable of having a $\boxed{V_{OH} = V_{DD}}$ and also results in a higher NM compared to saturated-load Inverter

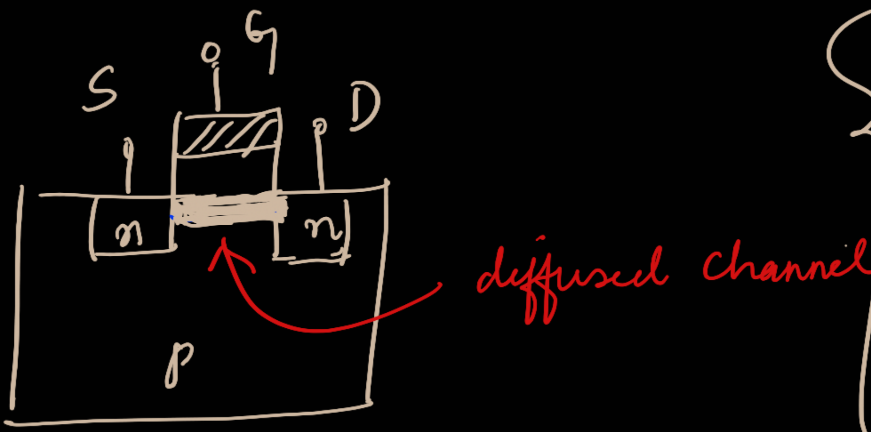
DISADVANTAGE:

(i) This configuration uses two separate power supply voltages.

(ii) Suffers from relatively high stand-by (DC) power dissipation.

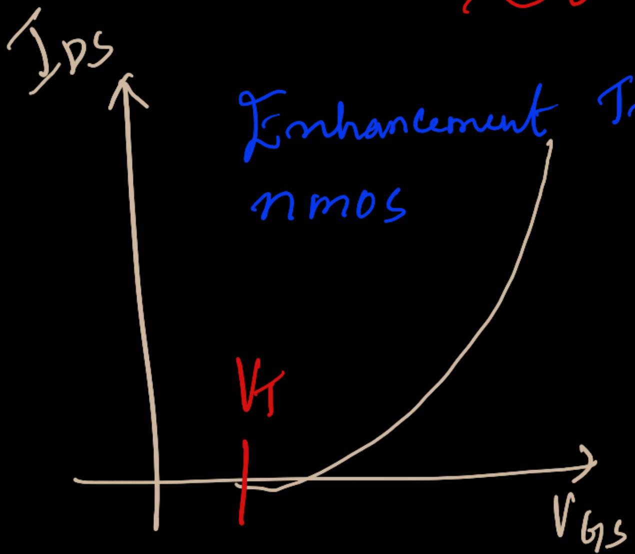
Enhancement load nmos inverters are [...] in any large scale digital not used [illegible] applications-

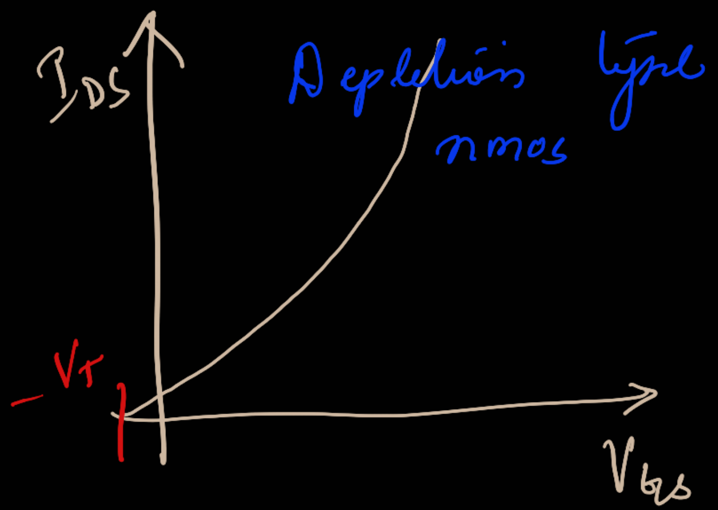
Depletion load n-MOS Inverters



If $V_{GS} = 0$, device is conducting with $I_D = I_{DO}$

Transfer Characteristics





* Depletion type nmos Conducts when $V_{GS} = 0$

* The gate and source is connected to make $V_{GS,load} = 0$ and the device operates in the linear or saturation region.

1. Threshold voltage of depletion load nmos

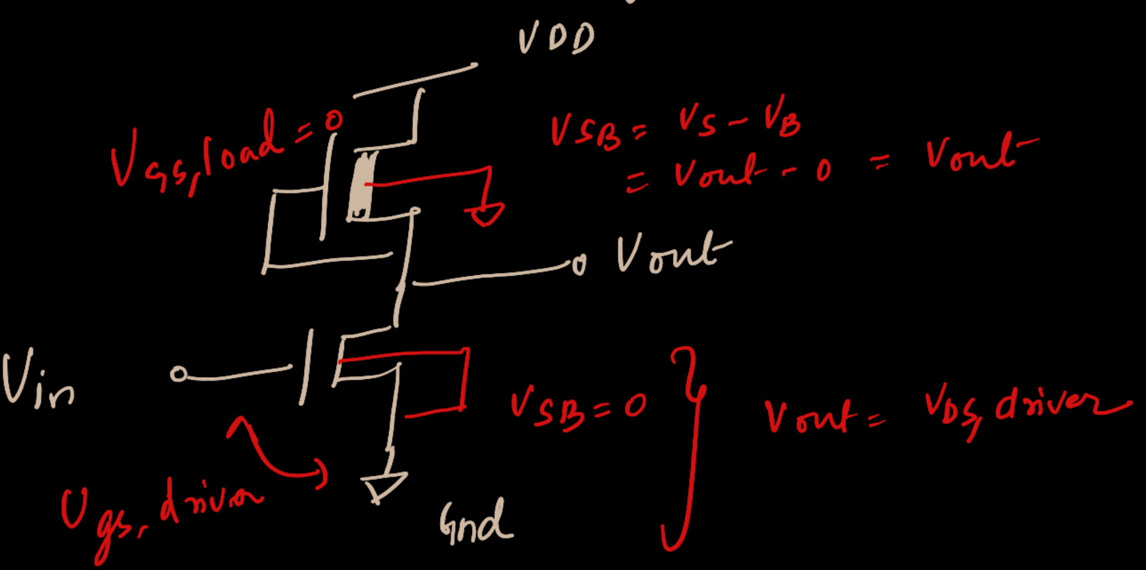
The threshold ...
is (-ve)

ie, $V_{to,load} < 0$ & $V_{to,driver} > 0$

* Fabrication process in depletion mode MOSFET requires additional processing steps, especially for the channel implant to adjust the threshold voltage of the load device

The immediate advantage of implementing this circuit configuration are —

* Sharp VTC and better noise margin
* Single power supply
* Small overall layout area.



* The gate and source of the load are connected

$$\boxed{V_{GS,load} = 0}$$

Hence, the load is always operating in Linear OR saturation region.

For the Load,

The threshold voltage value is a function of source substrate voltage $V_{SB} = V_{out}$

$$V_{T,load} = V_{T0} + \underbrace{\gamma \left(\sqrt{|2\phi_F| + V_{out}} - \sqrt{|2\phi_F|} \right)}_{\text{Extra term ie, body effect}}$$

Where

V_{T0} = Threshold voltage without body effect

γ = Body effect coefficient

$$= \frac{\sqrt{2 q N_a \varepsilon_s}}{C_{ox}}$$

ϕ_F = fermi potential of the material

$$|\phi_F| = \left(\frac{kT}{q}\right) \ln \left(\frac{N_a}{n_i}\right)$$

The operating mode of the load transistor is determined by the output voltage level.

(i) When the output voltage is small

$V_{out} < V_{DD} + V_{T,load}$ } nmos is in or

$V_{DS} > V_{gs} - V_T$

nmos Saturation

$$I_{D,load} = \frac{K_{n,load}}{2}\left(V_{GS} - V_{T,load}(v_{out})\right)^2$$

$V_{GS} = 0$

$$\boxed{I_{D,load} = \frac{K_{n,load}}{2}\left(-V_{T,load}(v_{out})\right)^2}$$

nmos Current in Saturation

For Large output voltages

$V_{out} > V_{DD} + V_{T,load}$

or

$V_{DS} < V_{GS} - V_{t,load}$

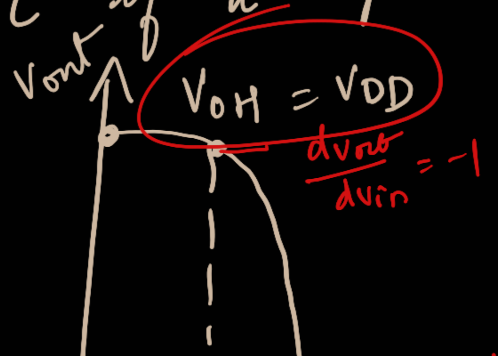
nmos is in the Linear region

$$I_{D,load} = \frac{K_n}{2}\left(2(V_{gs} - V_t)V_{ds} - V_{ds}^2\right)$$

$$\boxed{I_{D,load} = \frac{K_{n,load}}{2}\left[2\left|V_{T,load}(v_{out})\right|(V_{DD} - V_{out}) - (V_{DD} - V_{out})^2\right]}$$

nmos Current in Linear region.

VTC of a depletion load nmos inverter

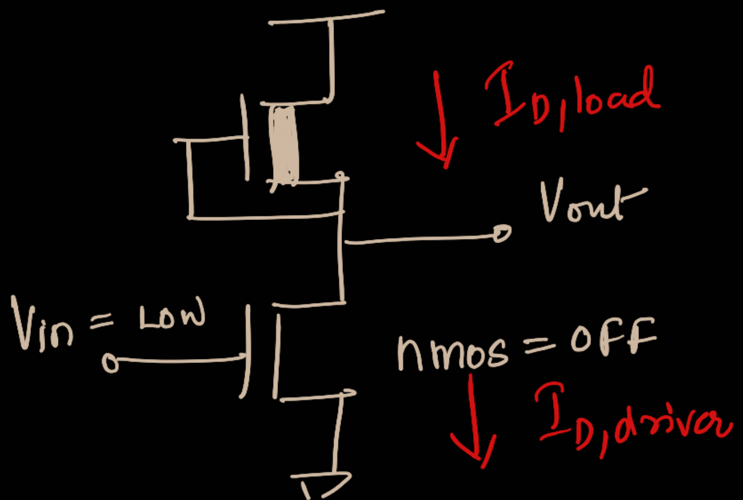


Power Calculation of Depletion load nmos Inverter:

$$\text{Power} = P_{DC} = V_{DD} \cdot I_D$$

If the duty cycle of the input pulse is 50%
then,

$$P_{DC} = \frac{V_{DD}}{2}\left[I_{DC}(V_{in} = \text{Low}) + I_{DC}(V_{in} = \text{High}) \right]$$



$$I_{DC}(V_{in} = \text{Low}) = 0$$

$$P_{DC} = \frac{V_{DD}}{2}\left[0 + I_{DC}(V_{in} = \text{High}) \right]$$

when

→ V_{in} = High, $V_{out} = V_{OL}$

→ Both driver and Load MOS are in the ON condition

$$\begin{cases} \text{~~Driver = Linear~~} \\ \text{Load = Saturation} \end{cases}$$

$$P_{DC} = \frac{V_{DD}}{2} \cdot \frac{k_{n,load}}{2} \left[\ldots - V_{T,load}(V_{OL}) \right]$$